RETAIL ANALYTICS / BUSINESS REPORT

Retail Sales Business Findings

A descriptive review of sales scale, concentration, holiday-week patterns, and markdown data quality.

AT A GLANCE

RECORDED WEEKLY SALES	OBSERVATIONS	STORES
6,737.22M	**421,570**	**45**
Total recorded weekly sales: 6,737,218,987.11	Store–department–week rows	Across 81 departments

HOLIDAY-WEEK AVERAGE	NON-HOLIDAY AVERAGE	AVERAGE UPLIFT
50.53M	**46.86M**	**7.84%**
10 holiday weeks	133 non-holiday weeks	Descriptive comparison

EXECUTIVE TAKEAWAY

Sales are concentrated in particular store types, stores, and departments. Average total weekly sales are higher in holiday weeks in this dataset, while substantial missingness in markdown fields limits promotion-related interpretation.

Analysis period: 5 February 2010 – 26 October 2012

Important: findings are descriptive summaries of historical patterns, not causal evidence.

01 / Executive Summary

This analysis covers 421,570 store-department-week observations across 45 stores, 81 departments, and 143 weekly dates from 2010-02-05 to 2012-10-26. Total recorded weekly sales are 6,737,218,987.11. Results show meaningful concentration in specific stores, store types, and departments, with higher average total weekly sales on holiday weeks than non-holiday weeks. These findings are descriptive summaries of historical patterns, not causal evidence.

Five quantified findings

01

Overall scale and average

Total weekly sales: 6,737,218,987.11; average sales per Store-Dept-Date row: 15,981.26.

02

Sales concentration by store

Store 20 is the largest single store by total sales at 301.40M (4.47% of total). The top 3 stores (20, 4, 14) together contribute 13.21% of total sales.

03

Store type mix

Type A: 4,331.01M (64.28%); Type B: 2,000.70M (29.70%); Type C: 405.50M (6.02%).

04

Holiday-week uplift

Average total sales per holiday week: 50.53M (10 weeks); non-holiday week: 46.86M (133 weeks). Holiday weeks are 7.84% higher on average.

05

Markdown field availability

Rows with all MarkDown1–5 missing: 270,138 (64.08%). Rows with any markdown present: 151,432 (35.92%). Sales from rows with any markdown present: 2,449.72M (36.36% of total).

Dataset scope and methodology

Item	Details
Source	Cleaned dataset at data/processed/retail_sales_clean.csv.
Observation grain	Store-Dept-Date (weekly); not individual customer transaction data.
SQL engine	DuckDB, with outputs exported to data/processed/analysis/.
Validation	Row count: 421,570; distinct Store-Dept-Date combinations: 421,570; total sales cross-check vs expected 6,737,218,987.11 within floating-point tolerance.
Aggregation	Weekly holiday comparison first aggregates total sales to Date, then compares holiday vs non-holiday weeks. No additional joins were introduced in the analysis step, so sales were not duplicated by join fan-out.

02 / Store & Department Performance

Sales are concentrated in a small group of high-contribution entities.

Top stores by total sales

Rank in supplied findings	Store	Total sales
1	Store 20	301.40M
2	Store 4	299.54M
3	Store 14	289.00M

Top departments by total sales

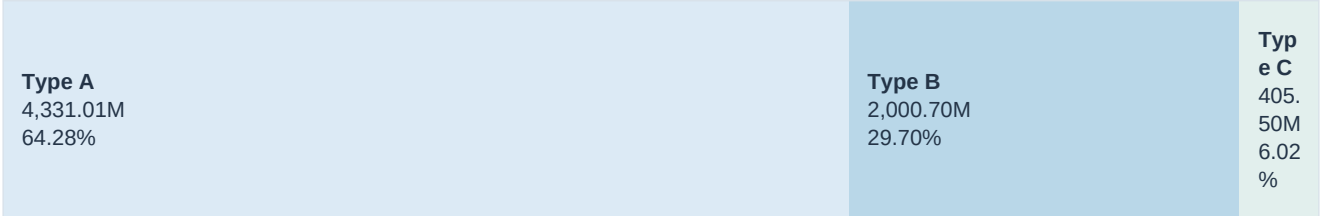
Department ID	Total sales
Dept 92	483.94M
Dept 95	449.32M
Dept 38	393.12M
Dept 72	305.73M
Dept 90	291.07M

Top Store–Department combinations

Store	Department	Total sales
Store 14	Dept 92	26.10M
Store 2	Dept 92	23.57M
Store 20	Dept 92	23.54M

Data note. Bottom combinations include negative totals (for example, Store 35, Dept 47: -3,567.18), consistent with net returns/adjustments in aggregate weekly records.

Store type mix



Department IDs are numeric codes; no department-name mapping was used.

03 / Holiday vs Non-Holiday Sales

The comparison was performed at the week level by first summing all store-department sales for each Date and then grouping by holiday flag.

HOLIDAY WEEKS	NON-HOLIDAY WEEKS	DIFFERENCE
10	**133**	**+7.84%**
Average total sales: 50.53M	Average total sales: 46.86M	Holiday-week average is higher

Weekly sales range

Measure	Non-holiday weeks	Holiday weeks
Number of weeks	133	10
Average total weekly sales	46.86M	50.53M
Minimum	39.60M	40.43M
Maximum	80.93M	66.59M

Interpretation

Holiday weeks show higher average total sales in this dataset. This is a descriptive association and does not by itself establish that holiday status causes the difference.

Markdown data limitations

MarkDown1–MarkDown5 are missing in a large share of rows, especially when considering all five fields together. Missing markdown values may reflect no promotion, unavailable data capture, or reporting gaps. The dataset alone cannot disambiguate these cases. Because of this ambiguity, markdown-related comparisons should be treated as exploratory and not interpreted as causal effects of promotions.

Markdown measure	Value
Rows with all MarkDown1–5 missing	270,138 (64.08%)
Rows with any markdown present	151,432 (35.92%)
Sales from rows with any markdown present	2,449.72M (36.36% of total)

04 / Practical Business Recommendations

Recommendations are framed as operational next steps consistent with the descriptive findings.

01

Prioritize high-contribution store types and entities

Use Type A stores and top-contributing stores/departments as primary pilots for assortment and labor planning scenarios.

02

Build a holiday-week operating playbook

Given higher average total sales on holiday weeks, pre-position staffing and inventory for identified holiday periods while monitoring volatility across weeks.

03

Improve markdown data capture standards

Require clearer markdown completeness definitions (for example, explicit zero vs unknown) to support reliable promotion-effect analysis.

05 / Limitations & Conclusion

Limitations

- Data grain is store-department-week, not customer transactions.
- Department IDs are numeric codes; no department-name mapping was used.
- Observational summaries do not control for confounders (seasonality, local factors, assortment changes, stockouts, etc.).
- Missing markdown fields constrain interpretation of promotion-related patterns.

Conclusion

The data indicates strong concentration of sales in specific store types, stores, and departments, with higher average weekly totals on holiday weeks. These results are useful for prioritization and planning, but causal claims require additional experimental or quasi-experimental analysis and improved promotion data quality.

Reporting standard: All figures and findings in this report are preserved from the supplied Markdown report. Visual formatting and section layout have been improved; no findings have been intentionally changed.